

逐句中英對照 全文翻譯

Integration of the SECI model and ChatGPT in higher education

SECI 模型與 ChatGPT 在高等教育的整合

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摘要 Abstract

In educational settings, the socialization, externalization, combination, and internalization (SECI) model promotes collaboration, the creation of new knowledge, and the integration of individual and collective experiences, thereby enriching learning with greater depth and meaning.

► 中文譯文 在教育情境中，社會化、外化、結合與內化（SECI）模型促進協作、新知識的創造，以及個人與集體經驗的整合，從而以更深的層次與意義豐富學習。

The objective of this study was to identify the factors that contribute to the optimal development of the SECI model through the use of ChatGPT, to understand their interaction.

► 中文譯文 本研究的目的，是辨識透過使用 ChatGPT 促成 SECI 模型最佳發展的因素，並理解這些因素之間的互動。

The experiment included undergraduate students of business sciences at public and private universities in Cusco, Huancayo, and Lima, Peru.

► 中文譯文 本實驗的對象，包含秘魯庫斯科、萬卡約與利馬等地公立與私立大學的商管科系大學部學生。

The students were divided into control and experimental groups, with 100 students in each group.

► 中文譯文 學生被分為控制組與實驗組，每組各 100 名學生。

The main selection criteria were prior knowledge of and willingness to use ChatGPT and being enrolled in a finance course.

► 中文譯文 主要的篩選標準為：先前對 ChatGPT 的認識與使用意願，以及正在修習一門財務課程。

The phases of the SECI model were evaluated using ChatGPT in the experimental group through group discussions, brainstorming sessions, the creation of a manual, and individual assessments, exploring the impact of ChatGPT usage on learning.

► 中文譯文 在實驗組中，透過小組討論、腦力激盪、編寫手冊與個別評量，使用 ChatGPT 來評估 SECI 模型的各個階段，藉以探討使用 ChatGPT 對學習的影響。

A comparison and description of each phase was conducted.

► 中文譯文 研究對各階段進行了比較與描述。

Results revealed that while the use of ChatGPT improved the quality of interaction and externalization of knowledge in the experimental group compared to the control group, it did not have a notable impact on the quality of combination or internalization of knowledge.

► 中文譯文 結果顯示，雖然使用 ChatGPT 相較於控制組改善了實驗組在知識互動與外化方面的品質，但對於知識的結合或內化品質並未帶來顯著影響。

A multinomial logistic regression analyzed the impact of ChatGPT usage in each phase.

► 中文譯文 一項多項式邏輯斯迴歸分析了在各階段使用 ChatGPT 的影響。

Findings showed that the SECI model without ChatGPT explained 63.4 % of the change in internalization, while with ChatGPT it was 41.3 %.

► 中文譯文 研究發現，未使用 ChatGPT 時，SECI 模型可解釋內化變化的 63.4%；而使用 ChatGPT 時則為 41.3%。

A positive correlation was observed between the use of ChatGPT and the performance in objective tests during the externalization and combination phases.

► 中文譯文 在外化與結合階段，觀察到使用 ChatGPT 與客觀測驗表現之間存在正相關。

However, while it enhanced the generation of ideas in the early stages of learning, it did not necessarily increase tacit knowledge in the internalization phase.

► 中文譯文 然而，雖然它在學習的早期階段強化了想法的產生，卻不必然提升內化階段的隱性知識。

The importance of using tools such as ChatGPT carefully and selectively, especially in the context of acquiring specialized knowledge, was emphasized.

► 中文譯文 研究強調，謹慎且有選擇地使用 ChatGPT 這類工具的重要性，特別是在取得專業知識的情境中。

1. 前言 1. Introduction

Artificial intelligence (AI) is rapidly transforming higher education by presenting new opportunities to enrich learning and address persistent challenges in teaching and knowledge management.

► 中文譯文 人工智慧 (AI) 正快速改變高等教育，為豐富學習、以及解決教學與知識管理中長期存在的挑戰，帶來新的機會。

Tools such as ChatGPT have shown the potential to improve active participation and personalize learning by processing large amounts of data and learning from its interactions with users [1,2].

► 中文譯文 ChatGPT 這類工具已展現出潛力，能藉由處理大量資料、並從與使用者的互動中學習，來提升主動參與並使學習個人化[1,2]。

However, this disruptive capability also brings limitations, such as generating superficial responses and inconsistency in texts [2], raising concerns about academic integrity and the ability of the app to develop complex and critical skills in students [1–3].

► 中文譯文 然而，這種顛覆性的能力也帶來限制，例如產生表淺的回應與文字前後不一致[2]，引發對學術誠信、以及該應用程式能否培養學生複雜與批判性技能的疑慮[1–3]。

In this context, the socialization, externalization, combination, and internalization (SECI) model, widely used in knowledge management, offers a conceptual framework to analyze the potential and challenges of integrating AI into teaching and learning processes.

► 中文譯文 在此脈絡下，廣泛應用於知識管理的社會化、外化、結合與內化（SECI）模型，提供了一個概念架構，用以分析將 AI 整合進教學與學習歷程的潛力與挑戰。

This model describes four phases: socialization, externalization, combination, and internalization that allow for the conversion of tacit knowledge into explicit knowledge and vice versa [4].

► 中文譯文 此模型描述四個階段：社會化、外化、結合與內化，使隱性知識與顯性知識能夠相互轉換[4]。

In the educational context, the SECI model has been a valuable tool to promote knowledge creation and transfer, facilitate collaboration between students and teachers, and promote innovative educational practices [5,6].

► 中文譯文 在教育情境中，SECI 模型一直是促進知識創造與傳遞、促進師生協作、並推動創新教育實踐的寶貴工具[5,6]。

Its application also improves the cocreation of knowledge and meaningful learning in the classroom [7].

► 中文譯文 其應用也改善了課堂中的知識共創與有意義的學習[7]。

Integration of ChatGPT into the SECI model has the potential to transform the educational experience by augmenting each of its phases.

► 中文譯文 將 ChatGPT 整合進 SECI 模型，具有透過強化各個階段來轉變教育體驗的潛力。

ChatGPT can support the socialization phase by acting as a moderator in group discussions, assist in externalization by organizing and clarifying the ideas generated by students, and contribute to combination by processing information from various sources to help structure complex concepts [8–10].

► 中文譯文 ChatGPT 可在社會化階段擔任小組討論的引導者來提供支援，在外化階段協助組織與釐清學生產生的想法，並在結合階段藉由處理來自不同來源的資訊以協助建構複雜概念[8–10]。

However, its effectiveness is not without challenges.

► 中文譯文 然而，它的成效並非毫無挑戰。

ChatGPT faces notable limitations in socialization, as it cannot capture nonverbal cues and cultural nuances, which are essential for effective tacit knowledge exchange and trust building [11].

► 中文譯文 ChatGPT 在社會化階段面臨明顯的限制，因為它無法捕捉非語言線索與文化上的細微差異，而這些對於有效的隱性知識交流與建立信任至關重要[11]。

In externalization, it is difficult for the app to interpret human subjectivities and translate them into explicit academic structures [12].

► 中文譯文 在外化階段，該應用程式難以詮釋人類的主觀感受，並將其轉化為明確的學術結構[12]。

During combination, its ability to synthesize and establish meaningful relations between diverse concepts is limited.

► 中文譯文 在結合階段，它綜合並建立不同概念之間有意義關聯的能力是有限的。

Finally, in internalization, while it facilitates access to and distribution of explicit knowledge, its ability to promote critical reflection and deep learning is limited [11].

► 中文譯文 最後，在內化階段，雖然它有助於取得與傳布顯性知識，但促進批判反思與深度學習的能力卻有限[11]。

The literature review reveals that the application of AI chatbots has focused primarily on information dissemination and response automation [13], which leaves a gap in their capacity to support the complete cycle of knowledge creation and transfer in the SECI model.

► 中文譯文 文獻回顧顯示，AI 聊天機器人的應用主要聚焦於資訊傳布與回應自動化[13]，這使得它們在支援 SECI 模型完整的知識創造與傳遞循環方面，留下了一個缺口。

In addition, the rapid evolution of conversational AI raises debates about its role in learning personalization, pedagogical design, and academic integrity [14,15].

► 中文譯文 此外，對話式 AI 的快速演進，引發了關於它在學習個人化、教學設計與學術誠信中所扮演角色的種種辯論[14,15]。

The gap between the current capabilities of AI and the pedagogical needs of higher education remains a critical area that requires further exploration.

► 中文譯文 AI 現有能力與高等教育教學需求之間的落差，仍是一個需要進一步探索的關鍵領域。

This study addresses a gap in the literature by exploring how ChatGPT can be integrated into the SECI model, specifically in teaching financial competencies in higher education.

► 中文譯文 本研究藉由探討如何將 ChatGPT 整合進 SECI 模型，特別是在高等教育的財務能力教學中，來填補文獻中的這一缺口。

Since these competencies require practical skills and critical analysis, ChatGPT could have a considerable impact on the learning of complex concepts and their practical applications.

► 中文譯文 由於這些能力需要實務技能與批判分析，ChatGPT 對於複雜概念的學習及其實務應用，可能具有相當的影響。

This study evaluates how ChatGPT facilitates each phase of the SECI model, acting as a moderator, idea facilitator, organizational assistant, and support for knowledge integration.

► 中文譯文 本研究評估 ChatGPT 如何促進 SECI 模型的各個階段，擔任引導者、想法促進者、組織助理，以及知識整合的支援。

This study is pioneering in investigating how ChatGPT can be integrated into each phase of the SECI model, an approach that has been little explored in previous research on AI and education.

► 中文譯文 本研究在探討如何將 ChatGPT 整合進 SECI 模型各階段方面具有開創性，這是先前關於 AI 與教育的研究中鮮少探討的取向。

Unlike general studies on the use of AI, this research provides a detailed empirical analysis of the specific contributions of ChatGPT to the socialization, externalization, and combination of knowledge, while observing the results obtained in the internalization phase and evaluating its potential and limitations for enhancing learning processes in higher education.

► 中文譯文 與一般關於 AI 使用的研究不同，本研究針對 ChatGPT 對知識的社會化、外化與結合的具體貢獻，提供了詳盡的實證分析，同時觀察內化階段所獲得的結果，並評估其對提升高等教育學習歷程的潛力與限制。

2. 文獻回顧 2. Literature review

The integration of AI tools such as ChatGPT into education has generated notable interest owing to its potential to transform traditional teaching models.

► 中文譯文 將 ChatGPT 這類 AI 工具整合進教育，因其轉變傳統教學模式的潛力而引起了顯著的關注。

The SECI model, which encompasses socialization, externalization, combination, and internalization of knowledge, serves as a framework to analyze how AI can influence the transfer and creation of knowledge in classrooms.

► 中文譯文 涵蓋知識社會化、外化、結合與內化的 SECI 模型，作為一個架構，用以分析 AI 如何影響課堂中知識的傳遞與創造。

This review explores the application of ChatGPT in each phase of the SECI model, highlighting its strengths and limitations.

► 中文譯文 本回顧探討 ChatGPT 在 SECI 模型各階段的應用，凸顯其優勢與限制。

社會化階段 Socialization Phase

In the socialization phase of the SECI model, individuals share tacit knowledge through interaction and experience [4].

► 中文譯文 在 SECI 模型的社會化階段，個體透過互動與經驗來分享隱性知識[4]。

ChatGPT facilitates and moderates group discussions, generating enthusiasm among students for its innovative features and adaptability [16,17].

► 中文譯文 ChatGPT 促進並引導小組討論，因其創新特性與適應性而在學生間激發熱情[16,17]。

It can guide debates, pose questions, and facilitate reflection and exchange of relevant experiences.

► 中文譯文 它能引導辯論、提出問題，並促進相關經驗的反思與交流。

However, ChatGPT has limitations.

► 中文譯文 然而，ChatGPT 有其限制。

It does not capture nonverbal cues such as body language or tone of voice, which are key elements for building trust and strengthening relations in tacit knowledge exchange [18].

► 中文譯文 它無法捕捉肢體語言或語調等非語言線索，而這些是隱性知識交流中建立信任與強化關係的關鍵元素[18]。

Moreover, it may struggle to grasp cultural nuances, limiting its ability to cultivate deep understanding among students.

► 中文譯文 此外，它可能難以掌握文化上的細微差異，從而限制其在學生間培養深度理解的能力。

While it facilitates dialogue, it cannot replace those human interactions necessary for collaborative learning [11].

► 中文譯文 雖然它促進對話，卻無法取代協作學習所必需的那些人際互動[11]。

外化階段 Externalization Phase

In the externalization phase, tacit knowledge is transformed into explicit knowledge [4].

► 中文譯文 在外化階段，隱性知識被轉化為顯性知識[4]。

ChatGPT is particularly useful at this stage, helping students to organize and structure their ideas.

► 中文譯文 ChatGPT 在此階段特別有用，能協助學生組織並建構其想法。

Its ability to generate real-time text and provide clear examples facilitates the expression of thoughts and the exchange of ideas [19].

► 中文譯文 它即時生成文字並提供清楚範例的能力，促進了想法的表達與交流[19]。

In activities such as brainstorming and mind mapping, ChatGPT acts as a facilitator, transforming abstract ideas into more concrete and coherent structures [20].

► 中文譯文 在腦力激盪與心智圖這類活動中，ChatGPT 擔任促進者，將抽象的想法轉化為更具體、更連貫的結構[20]。

However, concerns have been raised about potential dependency.

► 中文譯文 然而，已有人對潛在的依賴提出疑慮。

By offering immediate and organized responses, students may rely on ChatGPT instead of developing their own critical thinking processes [21].

► 中文譯文 由於提供即時且有條理的回應，學生可能依賴 ChatGPT，而非發展自身的批判思考歷程[21]。

This could hinder the development of essential cognitive skills such as critical reflection and independent evaluation [22].

► 中文譯文 這可能妨礙批判反思與獨立評估等重要認知技能的發展[22]。

Moreover, ChatGPT does not always interpret human subjectivities accurately, which can reduce its effectiveness in translating complex or emotionally charged thoughts into coherent academic structures [23].

► 中文譯文 此外，ChatGPT 並不總能準確詮釋人類的主觀感受，這可能降低它將複雜或帶有情緒的想法轉化為連貫學術結構的成效[23]。

結合階段 Combination Phase

In the combination phase, explicit knowledge is organized to form a broader body of knowledge [4].

► 中文譯文 在結合階段，顯性知識被加以組織，以形成更廣泛的知識體[4]。

ChatGPT can help students gather information from various sources and create coherent texts, which is particularly useful for complex tasks requiring the integration of multiple concepts [10].

► 中文譯文 ChatGPT 能協助學生從各種來源蒐集資訊並產生連貫的文本，這對於需要整合多個概念的複雜任務特別有用[10]。

However, ChatGPT faces challenges in areas requiring precision, such as mathematics and the exact sciences, where it has provided both correct and incorrect answers, affecting its reliability [23].

► 中文譯文 然而，ChatGPT 在需要精確性的領域（如數學與精確科學）面臨挑戰，在這些領域它曾提供正確與錯誤兼具的答案，影響其可靠性[23]。

In accounting evaluations, students outperformed ChatGPT considerably [24].

► 中文譯文 在會計類的評量中，學生的表現遠勝於 ChatGPT[24]。

This suggests that while ChatGPT helps organize information, it has limitations in synthesizing complex concepts and requires human oversight to avoid errors in knowledge consolidation.

► 中文譯文 這顯示，雖然 ChatGPT 有助於組織資訊，但它在綜合複雜概念方面有所限制，並需要人為監督以避免知識整合時的錯誤。

內化階段 Internalization Phase

In the internalization phase, students apply knowledge in their own context [4].

► 中文譯文 在內化階段，學生在自身的脈絡中應用知識[4]。

ChatGPT facilitates access to explicit information and helps to clarify concepts, which are useful in fields such as math education, where it assists with basic questions and problems [11,25].

► 中文譯文 ChatGPT 有助於取得顯性資訊並協助釐清概念，這在數學教育等領域很有用，它能協助處理基本的問題與題目[11,25]。

However, ChatGPT has limitations in promoting deep learning and critical reflection.

▶ 中文譯文 然而，ChatGPT 在促進深度學習與批判反思方面有其限制。

Although useful for memorization, it does not foster the development of metacognitive skills or critical analysis [25].

▶ 中文譯文 雖然對記憶有幫助，但它並不能培養後設認知技能或批判分析[25]。

Furthermore, in areas such as financial mathematics, ChatGPT has struggled to provide clear and accurate references, preventing the construction of new knowledge [26].

▶ 中文譯文 此外，在財務數學這類領域，ChatGPT 一直難以提供清楚而準確的參考依據，妨礙了新知識的建構[26]。

To improve precision in numerical contexts, alternatives such as ComputeGPT, which runs real-time code for responses, have been proposed [26].

▶ 中文譯文 為提升數值情境中的精確性，已有人提出 ComputeGPT 等替代方案，它能執行即時程式碼以產生回應[26]。

3. 材料與方法 3. Materials and methods

In this research, a "post-test" design was implemented, as shown in Fig. 1.

▶ 中文譯文 在本研究中，採用了一種「後測」(post-test) 設計，如圖 1 所示。

This design distinguishes between two groups: the experimental group and the control group.

▶ 中文譯文 此設計區分兩個組別：實驗組與控制組。

The experimental group is exposed to the stimulus or treatment, which in this case is the use of ChatGPT, followed by a measurement to evaluate the results of this intervention.

▶ 中文譯文 實驗組接受刺激或處置，在本研究中即為使用 ChatGPT，隨後進行一次測量以評估此介入的結果。

Meanwhile, the control group does not receive treatment but is also subjected to a measurement to compare the results without the influence of the stimulus.

▶ 中文譯文 同時，控制組不接受處置，但同樣接受一次測量，以比較在沒有刺激影響下的結果。

This design allows measurement and comparison of how the intervention (use of ChatGPT) affects the development of the socialization, externalization, and combination phases of the SECI model and subsequently evaluates the process of internalization of knowledge in both groups.

▶ 中文譯文 此設計能測量並比較介入（使用 ChatGPT）如何影響 SECI 模型中社會化、外化與結合階段的發展，並隨後評估兩組的知識內化歷程。

The structure of the "post-test" ensures that the intact groups can be assessed at the end of the experiment, allowing for the isolation of the specific effect of the intervention.

▶ 中文譯文 「後測」的結構確保完整的組別能在實驗結束時受到評估，從而能夠分離出介入的特定效果。

The research comprised 200 undergraduate students of management from various regions of Peru: 80 from a public university in Cusco, 50 from a private university in Huancayo, and 70 from a private university in Lima.

▶ 中文譯文 本研究包含 200 名來自秘魯不同地區的管理科系大學部學生：80 名來自庫斯科一所公立大學、50 名來自萬卡約一所私立大學、70 名來自利馬一所私立大學。

The selection of participants was based on their prior knowledge on ChatGPT, interest in its use, and enrollment in a finance course.

▶ 中文譯文 參與者的篩選，是根據他們先前對 ChatGPT 的認識、使用興趣，以及修習一門財務課程。

All students had previously completed courses related to the use of technology according to the curriculum of their programs.

▶ 中文譯文 所有學生先前都已依其課程的課表，完成與科技使用相關的課程。

Regarding financial knowledge, an entry exam was administered at the beginning of the course, demonstrating a uniform level of understanding in this area.

► 中文譯文 在財務知識方面，課程開始時施行了一次入學測驗，顯示出在此領域具有一致的理解水準。

The students were divided into two groups of 100 each: a control group and an experimental group.

► 中文譯文 學生被分為各 100 名的兩組：控制組與實驗組。

Fig. 2 illustrates the research framework, which began with a review of the literature to define the two main objectives.

► 中文譯文 圖 2 說明了研究架構，它始於文獻回顧以界定兩個主要目標。

Subsequently, an experiment was conducted based on the four phases of the SECI model: socialization (through group discussions), externalization (through brainstorming and mind mapping), combination (creating a manual), and internalization (objective evaluation of acquired knowledge).

► 中文譯文 隨後，依據 SECI 模型的四個階段進行一項實驗：社會化（透過小組討論）、外化（透過腦力激盪與心智圖）、結合（編寫手冊），以及內化（對已習得知識的客觀評量）。

This process allowed for the assessment of the impact of using ChatGPT on learning and knowledge application.

► 中文譯文 此歷程使我們能評估使用 ChatGPT 對學習與知識應用的影響。

To validate the measurement instruments, a reliability analysis was performed using McDonald's Omega technique, yielding values of 0.726 for the control group and 0.723 for the experimental group.

► 中文譯文 為驗證測量工具，使用 McDonald's Omega 技術進行了信度分析，控制組得到 0.726、實驗組得到 0.723 的數值。

Data were analyzed using two approaches: descriptive statistics to describe and compare differences between groups (Objective 1) and multinomial logistic regression analysis to explore the impact of the phases of the SECI model on learning (Objective 2).

► 中文譯文 資料以兩種取向進行分析：以描述統計來描述並比較組間差異（目標一），以及以多項式邏輯斯迴歸分析來探討 SECI 模型各階段對學習的影響（目標二）。

In this analysis, the phases of socialization, externalization, and combination were treated as independent variables, while internalization was the dependent variable.

► 中文譯文 在此分析中，社會化、外化與結合階段被視為自變項，而內化則為依變項。

Finally, the findings were synthesized into contributions, limitations, and recommendations, providing a comprehensive conclusion to the research.

► 中文譯文 最後，研究發現被綜合為貢獻、限制與建議，為本研究提供一個完整的結論。

3.1 ChatGPT 的使用 3.1 Use of ChatGPT

ChatGPT was chosen for its widespread adoption in the academic field and its ability to generate detailed and personalized responses, which facilitate learning and supports research tasks.

► 中文譯文 選擇 ChatGPT，是因為它在學術領域廣泛被採用，且能產生詳盡而個人化的回應，從而促進學習並支援研究任務。

Unlike other chatbots, it offers adaptive and fluid responses, allowing for more effective tutoring.

► 中文譯文 與其他聊天機器人不同，它提供具適應性且流暢的回應，使輔導更為有效。

Its accessibility and free availability make it especially useful for students.

► 中文譯文 它的可及性與免費取得，使其對學生特別有用。

In the SECI model, the capabilities of ChatGPT are integrated into the phases of socialization, externalization, and combination.

► 中文譯文 在 SECI 模型中，ChatGPT 的能力被整合進社會化、外化與結合階段。

In the socialization phase, ChatGPT moderates discussions, simulating conversations that promote the exchange of experiences and tacit knowledge.

▶ 中文譯文 在社會化階段，ChatGPT 引導討論，模擬促進經驗與隱性知識交流的對話。

During externalization, it guides brainstorming sessions and the creation of concept maps, helping to organize complex concepts and providing clear examples to effectively codify ideas.

▶ 中文譯文 在外化階段，它引導腦力激盪與概念圖的建立，協助組織複雜概念，並提供清楚範例以有效編碼想法。

In combination, ChatGPT acts as a specialized assistant, collecting and systematizing information, allowing students to create detailed materials and structure content coherently, facilitating the consolidation of explicit knowledge.

▶ 中文譯文 在結合階段，ChatGPT 擔任專門助理，蒐集並系統化資訊，讓學生能建立詳細的材料並連貫地組織內容，促進顯性知識的整合。

Version 3.5 selected for this study offers advantages such as rapid information retrieval and support in academic writing.

▶ 中文譯文 本研究所選用的 3.5 版本，提供了快速資訊檢索與學術寫作支援等優勢。

However, it may have limitations in specialized critical analysis.

▶ 中文譯文 然而，它在專業的批判分析方面可能有所限制。



Fig. 1. The experimental design involves manipulating the presence or absence of the independent variable with the participation of an experimental group and a control group. An exclusive post-test design is used employing intact groups.

圖 1. 此實驗設計操弄自變項的有無，並由實驗組與控制組共同參與；採用以完整組別進行的純後測設計。（圖例：G = 參與學生的分組；X = G1 組使用 ChatGPT 作為教學工具；O = 財務能力的學業表現評量；- = 控制組 G2 不使用 ChatGPT。）

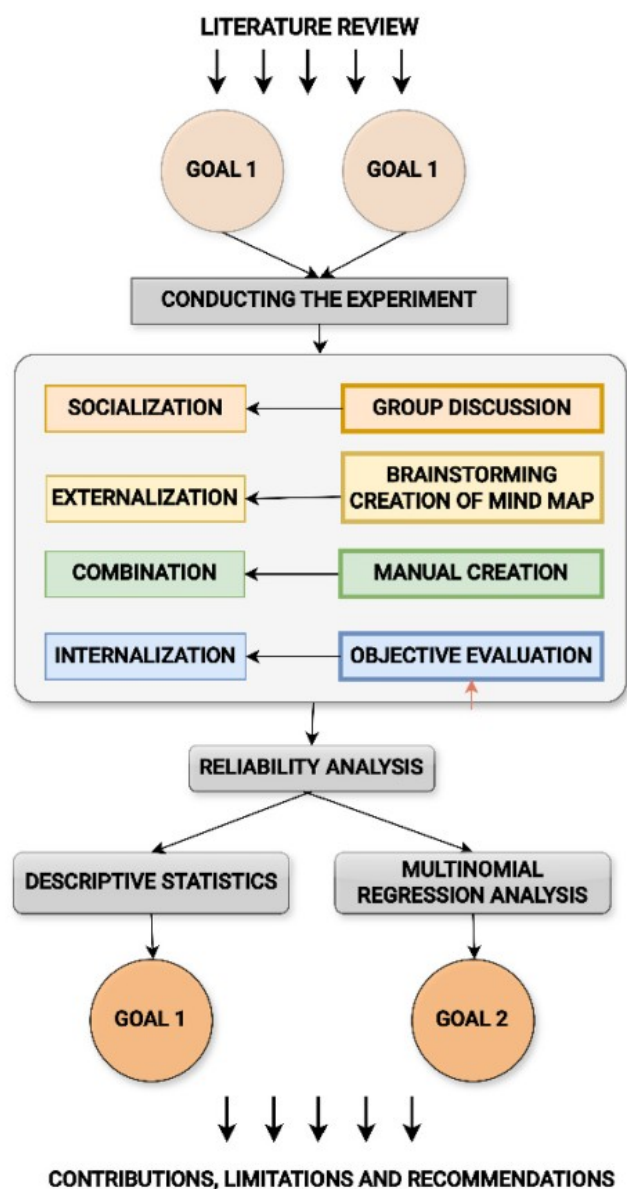


Fig. 2. Research framework.

Fig. 2. Research framework.

圖 2. 研究架構。

3.2 實驗的開展 3.2 Development of the experiment

For the development of the experiment, the activities and the main objectives of each phase of the SECI model were defined, as shown in Table 1.

► 中文譯文 為開展實驗，研究界定了 SECI 模型各階段的活動與主要目標，如表 1 所示。

In these activities, participants discussed, organized, and applied financial concepts related to amortization.

► 中文譯文 在這些活動中，參與者討論、組織並應用與攤銷相關的財務概念。

The phases were designed to facilitate the exchange, codification, consolidation, and practical evaluation of knowledge.

► 中文譯文 這些階段的設計，旨在促進知識的交流、編碼、整合與實務評量。

表 1. 實驗中 SECI 模型各階段的活動與目標 Table 1.

階段 Phase	活動 Activity	主要目標 Main Objective
Socialization 社會化	Group discussion on financial amortization (20 min). Groups of four students share their experience and knowledge. 針對財務攤銷的小組討論（20 分鐘）。四人一組分享經驗與知識。	Facilitate the exchange of tacit knowledge and improve the quality of group interaction. 促進隱性知識的交流，並改善小組互動的品質。
Externalization 外化	1) Brainstorming session (10 min) to generate concepts about financial amortization. 2) Creation of mind maps (20 min) to organize concepts visually, integrating ChatGPT. 1) 腦力激盪（10 分鐘）以產生財務攤銷的概念。2) 建立心智圖（20 分鐘）以視覺化方式組織概念，並整合 ChatGPT。	Encourage the expression and codification of ideas and organize explicit knowledge in a structured manner. 鼓勵想法的表達與編碼，並以結構化方式組織顯性知識。
Combination 結合	Creation of a detailed manual on financial amortization (30 min), including basic concepts, calculations, and applied examples. 編寫一份詳細的財務攤銷手冊（30 分鐘），包含基本概念、計算與應用範例。	Consolidate and expand explicit knowledge, enhancing the ability to synthesize and apply financial concepts. 整合並擴展顯性知識，提升綜合與應用財務概念的能力。
Internalization 內化	Objective evaluation (40 min) through a written test with eight questions (theoretical and calculation based) on amortization. 透過一份含八題（理論與計算題）的攤銷筆試進行客觀評量（40 分鐘）。	Assess the degree of internalization and application of acquired financial knowledge in theoretical and practical contexts. 評估已習得財務知識在理論與實務情境中的內化與應用程度。

Table 2 highlights the differences between the control group and the experimental group, focusing on the integration of ChatGPT.

► 中文譯文 表 2 凸顯了控制組與實驗組之間的差異，聚焦於 ChatGPT 的整合。

This comparison indicates how the phases of the SECI model are developed: the control group follows a traditional, nontechnological approach, while the experimental group incorporates ChatGPT to enhance discussion, structure ideas, and create materials on financial amortization.

► 中文譯文 此比較指出 SECI 模型各階段如何開展：控制組採用傳統、非科技的取向，而實驗組則納入 ChatGPT 來強化討論、建構想法，並製作財務攤銷的材料。

The table specifically details how ChatGPT is integrated into each phase, optimizing the socialization, externalization, and combination of knowledge.

► 中文譯文 此表具體詳述 ChatGPT 如何整合進各階段，以最佳化知識的社會化、外化與結合。

Fig. 3 illustrates the four phases of the SECI model (socialization, externalization, combination, and internalization) in the context of learning about financial amortization.

► 中文譯文 圖 3 以財務攤銷的學習情境，呈現 SECI 模型的四個階段（社會化、外化、結合與內化）。

Each phase is associated with a specific activity: an informal discussion (socialization), brainstorming and mind map creation (externalization), manual preparation (combination), and individual objective evaluation (internalization).

► 中文譯文 每個階段都對應一項特定活動：非正式討論（社會化）、腦力激盪與心智圖建立（外化）、手冊編寫（結合），以及個別客觀評量（內化）。

The continuous flow of knowledge is highlighted, moving from interaction and dialogue to structuring and practical application, promoting active learning and the integration of explicit knowledge.

► 中文譯文 此圖凸顯知識的連續流動，從互動與對話走向結構化與實務應用，促進主動學習與顯性知識的整合。

3.3 變項的控制與提示語的使用 3.3 Control of variables and use of prompts

To ensure consistency in the use of ChatGPT and minimize the influence of external variables, such as differences in instructions, components were established for developing prompts for the experimental group.

► 中文譯文 為確保使用 ChatGPT 的一致性，並將指令差異等外部變項的影響降到最低，研究為實驗組建立了開發提示語的組成要素。

These components were carefully designed to guide interactions with ChatGPT during each phase of the process, providing context, role, and specific tasks for the chatbot, as detailed in Table 3.

► 中文譯文 這些要素經過審慎設計，用以在歷程的各階段引導與 ChatGPT 的互動，為聊天機器人提供脈絡、角色與特定任務，詳如表 3 所示。

This approach ensures clarity, alignment with objectives, and uniformity in the interactions.

► 中文譯文 此取向確保了互動的清晰、與目標的一致，以及一致性。

With the essential components that the prompts needed to include in place, participants carefully structured the prompts to align the context, assigned role, and specific tasks.

► 中文譯文 在提示語所需包含的基本要素就緒後，參與者審慎地建構提示語，以對齊脈絡、所指派的角色與特定任務。

For example, during the socialization phase, prompts were developed with the following content: "You are a group discussion moderator (Role) on financial amortization (Context). Generate open-ended and specific questions that address basic concepts, personal experiences, and practical applications to foster dialogue (Task)."

► 中文譯文 例如，在社會化階段，提示語以如下內容開發：「你是一位財務攤銷（脈絡）的小組討論引導者（角色）。請產生開放式且具體的問題，涵蓋基本概念、個人經驗與實務應用，以促進對話（任務）。」

Similarly, during the externalization phase, prompts like the following were implemented: "You are a knowledge transformation facilitator (Role) (Context). Help participants generate key ideas about financial amortization and organize structured concepts with clear and practical examples (Task)."

► 中文譯文 同樣地，在外化階段，採用了如下的提示語：「你是一位知識轉化的促進者（角色）（脈絡）。請協助參與者產生關於財務攤銷的關鍵想法，並以清楚而實用的範例組織結構化的概念（任務）。」

3.4 各階段的評估準則 3.4 Evaluation criteria for the phases

Specific evaluation criteria were established for each phase of the SECI model, aimed at measuring student participation and interaction with both content and technology, from the transfer of tacit knowledge to the consolidation of explicit knowledge.

► 中文譯文 研究為 SECI 模型各階段建立了特定的評估準則，旨在測量學生對內容與科技的參與及互動，涵蓋從隱性知識的傳遞到顯性知識的整合。

Table 4 summarizes these criteria, which reflect key aspects of the cognitive and social activities of the participants, including the use of ChatGPT.

► 中文譯文 表 4 彙整了這些準則，它們反映了參與者認知與社會活動的關鍵面向，包含 ChatGPT 的使用。

In the internalization phase, which is not included in the table, knowledge assimilation was assessed through an objective test comprising eight questions, graded on a Likert scale from 1 to 5.

► 中文譯文 至於未列入表中的內化階段，知識的吸收是透過一份含八題、以 1 到 5 分李克特量尺計分的客觀測驗來評估。

4. 結果與討論 4. Results and discussion

4.1 結果 4.1 Results

Fig. 4 presents the results of the socialization phase, where the students exchanged knowledge about financial amortization.

► 中文譯文 圖 4 呈現社會化階段的結果，學生在此交流了關於財務攤銷的知識。

In the control group, most of the students fell into low to moderate levels of interaction, with 38 % in the "low" category and 40 % in "moderate," while only 22 % reached "good" or "excellent" levels.

► 中文譯文 在控制組中，多數學生落在低至中等的互動水準，38%屬「低」、40%屬「中等」，僅22%達到「良好」或「優異」水準。

In contrast, the experimental group, which used ChatGPT, showed a more balanced distribution toward higher levels, with 52 % in the "moderate," 33 % in the "good," and 3 % in the "excellent" categories.

► 中文譯文 相對地，使用 ChatGPT 的實驗組呈現出朝較高水準較為平衡的分布，52%屬「中等」、33%屬「良好」、3%屬「優異」。

These results indicate that the inclusion of ChatGPT in the socialization process improved the quality and level of student interaction compared with the control group.

► 中文譯文 這些結果顯示，在社會化歷程中納入 ChatGPT，相較於控制組改善了學生互動的品質與水準。

Fig. 5 shows the results of the externalization phase, where students participated in brainstorming sessions and created mind maps on financial amortization.

► 中文譯文 圖 5 呈現外化階段的結果，學生在此參與腦力激盪並建立財務攤銷的心智圖。

In the control group, 24 % of the students showed a low level of externalization, while 47 % reached a "good" level.

► 中文譯文 在控制組中，24%的學生展現低度外化，而 47%達到「良好」水準。

Only 15 % achieved an "excellent" level.

► 中文譯文 僅 15%達到「優異」水準。

In contrast, the experimental group, which used ChatGPT, had no students in the "low" or "moderate" levels.

► 中文譯文 相對地，使用 ChatGPT 的實驗組沒有任何學生落在「低」或「中等」水準。

Most students reached higher levels, with 36 % achieving a "good" level and 64 % an "excellent" level.

► 中文譯文 多數學生達到較高水準，36%達「良好」、64%達「優異」。

These results suggest that the use of ChatGPT facilitated a higher quality of externalization, strengthening its potential to assist in the expression and codification of tacit knowledge during collaborative activities such as idea generation and mind map creation.

► 中文譯文 這些結果顯示，使用 ChatGPT 促成了更高品質的外化，強化了它在想法產生與心智圖建立等協作活動中，協助表達與編碼隱性知識的潛力。

表 2. 控制組與實驗組在 SECI 模型各階段的活動比較 Table 2.

階段 Phase	控制組 Control Group	實驗組 (含 ChatGPT) Experimental Group
Socialization 社會化	Students engage in a free discussion, without the use of technology, based solely on verbal exchange. 學生進行自由討論，不使用科技，僅依賴口語交流。	ChatGPT is used to guide conversation, enrich the exchange of ideas, and facilitate reflection on experiences related to financial amortization. 使用 ChatGPT 引導對話、豐富想法交流，並促進對財務攤銷相關經驗的反思。
Externalization 外化	1) Ideas are generated verbally without technological tools. 2) Mind maps are created manually based on group discussions. 1) 想法以口語產生，不使用科技工具。2) 依據小組討論手動建立心智圖。	1) ChatGPT assists in the generation and organization of ideas during brainstorming. 2) ChatGPT helps create mind maps by helping to create visual structures and clarification of concepts. 1) ChatGPT 在腦力激盪時協助想法的產生與組織。2) ChatGPT 協助建立心智圖，幫忙建立視覺結構並釐清概念。
Combination 結合	Students create the manual without the support of ChatGPT, using only class materials and prior discussions. 學生不借助 ChatGPT 編寫手冊，僅使用課堂材料與先前的討論。	ChatGPT is used to generate detailed and varied examples, verify mathematical calculations, and enrich the manual with FAQs and practical tips. 使用 ChatGPT 產生詳細而多樣的範例、

階段 Phase	控制組 Control Group	實驗組 (含 ChatGPT) Experimental Group
		驗證數學計算，並以常見問答與實用建議豐富手冊。
Internalization 內化	The evaluation is conducted without the intervention of ChatGPT, measuring the knowledge acquired from previous activities. 評量在沒有 ChatGPT 介入下進行，測量先前活動所習得的知識。	Although ChatGPT is not used in this phase, the evaluation assesses the understanding and application of the knowledge supported by ChatGPT in earlier phases. 雖然此階段不使用 ChatGPT，但評量檢視的是在先前階段由 ChatGPT 支援所習得知識的理解與應用。

Fig. 6 shows the results of the combination phase, in which the students created a financial amortization manual.

► 中文譯文 圖 6 呈現結合階段的結果，學生在此編寫了財務攤銷手冊。

In the control group, 47 % of the students reached a "moderate" level and 52 % achieved a "good" level.

► 中文譯文 在控制組中，47%的學生達到「中等」水準、52%達到「良好」水準。

The experimental group, which used ChatGPT, showed similar results, with 40 % at the "moderate" level and 47 % at the "good" level, reflecting an almost identical distribution to the control group.

► 中文譯文 使用 ChatGPT 的實驗組呈現相似的結果，40%屬「中等」、47%屬「良好」，與控制組的分布幾乎相同。

Both groups had minimal representation at the "low" and "excellent" levels.

► 中文譯文 兩組在「低」與「優異」水準的占比都很少。

These results indicate that the use of ChatGPT did not have a notable impact on the quality of the combination phase, as the experimental and control groups similarly consolidated the explicit knowledge generated during externalization to create a comprehensive and organized body of knowledge.

► 中文譯文 這些結果顯示，使用 ChatGPT 對結合階段的品質並未帶來顯著影響，因為實驗組與控制組同樣地整合了外化階段所產生的顯性知識，形成一個完整而有組織的知識體。

Fig. 7 shows the results of the internalization phase, where an individual objective evaluation was carried out for both groups.

► 中文譯文 圖 7 呈現內化階段的結果，兩組都進行了個別的客觀評量。

In the control group, 57 % of the students reached a "moderate" level, while 43 % reached a "low" level.

► 中文譯文 在控制組中，57%的學生達到「中等」水準，而 43%達到「低」水準。

However, the experimental group, which used ChatGPT, presented a similar distribution, with 57 % at the "low" level and 35 % at the "moderate" level, with no students in the "good" or "excellent" levels.

► 中文譯文 然而，使用 ChatGPT 的實驗組呈現相似的分布，57%屬「低」、35%屬「中等」，沒有學生落在「良好」或「優異」水準。

These results indicate that there were no notable differences in knowledge internalization between the control group and the experimental group.

► 中文譯文 這些結果顯示，控制組與實驗組在知識內化上並無顯著差異。

Both groups integrated the acquired knowledge as part of their skills and personal understanding, although the control group demonstrated a slight advantage in internalization quality compared with the experimental group.

► 中文譯文 兩組都將習得的知識整合為自身技能與個人理解的一部分，不過控制組在內化品質上相較實驗組展現出些微的優勢。

Multinomial logistic regression was chosen to analyze the effect of ChatGPT owing to its ability to handle categorical variables and predict the probability of belonging to each level.

► 中文譯文 之所以選擇多項式邏輯斯迴歸來分析 ChatGPT 的效果，是因為它能處理類別變項並預測屬於各個水準的機率。

This allows a comparison of how ChatGPT impacts the phases of the SECI model.

► 中文譯文 這使我們能比較 ChatGPT 如何影響 SECI 模型的各階段。

The technique helps identify patterns and relations between the independent variables (socialization, externalization, and combination) and the dependent variable (internalization), providing a detailed analysis of the effect of ChatGPT on learning.

► 中文譯文 此技術有助於辨識自變項（社會化、外化與結合）與依變項（內化）之間的型態與關聯，提供 ChatGPT 對學習效果的詳盡分析。

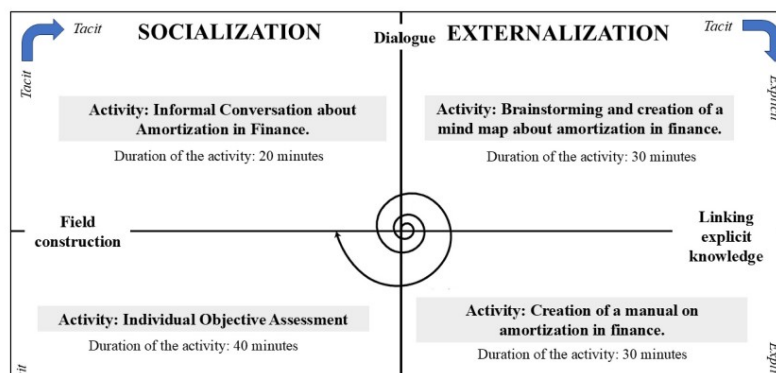


Fig. 3. Stages of the SECI model applied in the experiment.

圖 3. 實驗中所應用的 SECI 模型各階段。

表 3. 社會化、外化與結合階段的提示語組成要素 Table 3.

階段 Phase	脈絡 Context	角色 Role	任務 Tasks
Socialization 社會化	Group discussion on financial amortization 財務攤銷的小組討論	Group discussion moderator 小組討論引導者	Generate questions to stimulate dialogue. Facilitate conversation among participants. Guide the exchange of experience and knowledge. 產生問題以激發對話。促進參與者之間的交談。引導經驗與知識的交流。
Externalization 外化	Brainstorming and creation of conceptual maps on amortization 關於攤銷的腦力激盪與概念圖建立	Facilitator of knowledge transformation 知識轉化的促進者	Guide the brainstorming session. Help organize and structure concepts. Provide clear examples to illustrate ideas. 引導腦力激盪。協助組織並建構概念。提供清楚範例以說明想法。
Combination 結合	Creation of a detailed manual on financial amortization 編寫詳細的財務攤銷手冊	Specialized assistant in information gathering 資訊蒐集的專門助理	Gather data from various sources. Summarize key points. Structure the information in a coherent and clear manner. 從各種來源蒐集資料。摘要重點。以連貫而清楚的方式組織資訊。

Table 5 shows that the final model fits better than the initial beta value, supported by an AIC index of 70.436 and a significance level below 0.05, which confirms the fit of the multinomial regression model.

► 中文譯文 表 5 顯示最終模型的適配優於初始的 beta 值，AIC 指標為 70.436、顯著性低於 0.05，證實了多項式迴歸模型的適配良好。

In Table 6, the values obtained in the pseudo-R-squared are shown, selecting the highest one, which is Nagelkerke's with a coefficient of 0.634.

► 中文譯文 表 6 呈現各項擬 R 平方 (pseudo-R-squared) 的數值，取其中最高者，即 Nagelkerke 值，係數為 0.634。

This indicates that the final model explains 63.4 % of the variance in the dependent variable (internalization), which is a notable value for measuring causality.

► 中文譯文 這代表最終模型能解釋依變項（內化）63.4%的變異，就衡量因果關係而言是個相當顯著的數值。

In Table 7, it can be seen that in the SECI model, the socialization factor does not predict the internalization score because its p-value of 0.916 is not statistically significant.

► 中文譯文 從表 7 可見，在 SECI 模型中，社會化因子並不能預測內化分數，因為其 p 值為 0.916，未達統計顯著。

However, according to the coefficient Exp (B), as socialization increases by one unit, the student is 1.032 times more likely to obtain an intermediate score in the objective test conducted for internalization.

► 中文譯文 不過，依據 Exp(B)係數，社會化每增加一個單位，學生在內化的客觀測驗中取得中等分數的可能性會提高為 1.032 倍。

The externalization factor predicts the score obtained in internalization, as its p-value of 0.000 is statistically significant.

► 中文譯文 外化因子能預測內化所得的分數，因為其 p 值為 0.000，達到統計顯著。

Furthermore, the coefficient of Exp (B) or the odds ratio indicates that as the externalization increases by one unit, the student is 3.488 times more likely to obtain an intermediate score in the objective test conducted for internalization.

► 中文譯文 此外，Exp(B)係數（即勝算比）顯示，外化每增加一個單位，學生在內化客觀測驗中取得中等分數的可能性會提高為 3.488 倍。

Similarly, the combination factor predicts the score obtained in internalization, since its p-value of 0.006 is statistically significant.

► 中文譯文 同樣地，結合因子也能預測內化所得的分數，因為其 p 值為 0.006，達到統計顯著。

However, the coefficient shows that as the combination increases by one unit, the student is 0.269 times more likely to obtain a basic score in the objective test conducted for internalization, which is the lowest of the three independent variables.

► 中文譯文 然而係數顯示，結合每增加一個單位，學生在內化客觀測驗中取得基礎分數的可能性為 0.269 倍，是三個自變項中最低的。

表 4. 社會化、外化與結合階段的評估準則 Table 4.

階段 Phase	評估準則 Criterion	描述 Description	作者 Author
Socialization 社會化	Active participation 主動參與	The participant actively and significantly contributes to the exchange of knowledge. 參與者主動且顯著地對知識交流做出貢獻。	[27]
	Facilitation of dialogue 促進對話	The participant helps to create an environment conducive to dialogue and interaction among the participants. 參與者協助營造有利於參與者間對話與互動的環境。	[28]
	Sharing relevant experiences 分享相關經驗	The participant shares experiences, values, beliefs, and personal emotions relevant to the topic. 參與者分享與主題相關的經驗、價值、信念與個人情感。	[29]
	Building relationships	The participant builds	[27]

階段 Phase	評估準則 Criterion	描述 Description	作者 Author
	建立關係	meaningful relationships during the socialization process. 參與者在社會化歷程中建立有意義的關係。	
	Development of Collective Understanding 發展集體理解	The participant contributes to the development of a collective understanding of knowledge within the group. 參與者促進組內對知識的集體理解之發展。	[27]
Externalization 外化	Inclusive and nonthreatening environment 包容且無威脅的環境	In the group, the participant appears confident in expressing their ideas freely. 在組內，參與者顯得有信心自由表達自己的想法。	[30]
	Recording of all relevant ideas 記錄所有相關想法	The participant takes notes and records all ideas related to the topic, regardless of their tangentiality. 參與者記下並記錄所有與主題相關的想法，無論其是否離題。	[30–32]
	Activation of prior knowledge 啟動先備知識	The participant activates previous knowledge and skills to evaluate existing information and seek new information. 參與者啟動先前的知識與技能，以評估既有資訊並尋求新資訊。	[30,33]
	Utilization of stimulating techniques 運用激發技巧	The participant uses techniques that encourage participation. 參與者運用能鼓勵參與的技巧。	[30,34]
	Adaptability and versatility 適應力與彈性	The participant adapts to and participates naturally in the brainstorming technique. 參與者能適應腦力激盪技巧並自然參與。	[35,36]
	Central concept 中心概念	The central concept is clear, relevant, and well-defined, guiding the development of the mind map effectively. 中心概念清楚、相關且明確界定，能有效引導心智圖的發展。	[37,38]
	Main topics and branching 主要主題與分支	The main topics and branches are defined clearly, organized hierarchically, and related effectively to the central concept. 主要主題與分支界定清楚、層次分明，並與中心概念有效連結。	[37]
	Keywords and minimalism 關鍵字與精簡	All branches are labeled with relevant and concise keywords that improve understanding of the mind map. 所有分支都以相關而精簡的關鍵字標示，以提升對心智圖的理解。	[39,40]
	Color coding and visual elements 色彩編碼與視覺元素	Color coding and visual elements are used creatively and coherently, significantly	[41]

階段 Phase	評估準則 Criterion	描述 Description	作者 Author
		enhancing the understanding and visual appeal of the mind map. 色彩編碼與視覺元素運用得具創意且連貫，顯著提升心智圖的理解度與視覺吸引力。	
	Links and relationships 連結與關聯	Clear and meaningful links are established between all branches, demonstrating a deep understanding of conceptual relations within the mind map. 所有分支之間建立清楚而有意義的連結，展現對心智圖內概念關聯的深度理解。	[42]
Combination 結合	Clarity and organization 清晰與組織	The manual is structured logically and the information is presented clearly. 手冊結構合乎邏輯，且資訊呈現清楚。	[43]
	Accuracy and reliability 準確與可靠	The information provided is accurate and reliable, supported by appropriate sources. 所提供的資訊準確而可靠，並有適當來源佐證。	[43]
	Relevance and pertinence 相關與切題	The content of the manual is relevant to the target audience. 手冊的內容與目標讀者相關。	[43]
	Usability and accessibility 易用與易取得	The manual is easy to use and access, with an intuitive design, a clear index, and an appropriate visual presentation. 手冊易於使用與取得，具直覺的設計、清楚的索引與適當的視覺呈現。	[43]
	Effectiveness in knowledge transmission 知識傳遞的成效	The manual conveys knowledge effectively, facilitating the reader's learning and understanding of the topic addressed. 手冊有效傳達知識，促進讀者對所談主題的學習與理解。	[43]

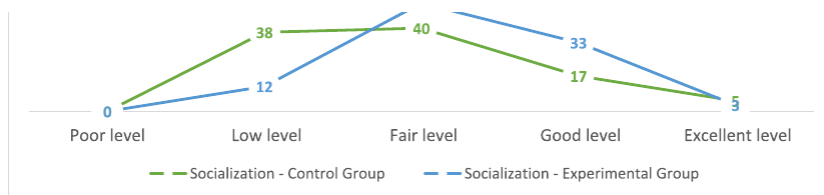


Fig. 4. Level of socialization.

id the dependent variable (internalization), providing a detailed analysis of the effect of ChatGPT on learning. Table 5 shows that the final model fits better than the initial beta value, supported by an AIC index of 70.436 and a significance level below 0.05, which confirms the fit of the multinomial regression model.

In Table 6, the values obtained in the pseudo-R-squared are shown, selecting the highest one, which is Nagelkerke's with a c

Fig. 4. Level of socialization.

圖 4. 社會化的水準。

註：a. 參考類別為「基礎」（Basic）。

From this point on, the results obtained in the experimental group are described using socialization, externalization, and combination as independent variables and internalization as a dependent variable.

▶ 中文譯文 從這裡開始，描述實驗組所得的結果，同樣以社會化、外化與結合作為自變項，內化作為依變項。

In Table 8, it is observed that the final model fits better than having only the beta value.

▶ 中文譯文 從表 8 可觀察到，最終模型的適配優於僅有 beta 值的情況。

This is demonstrated by the AIC index (96.713), which is lower compared to a significance of less than 0.05.

▶ 中文譯文 這由 AIC 指標（96.713）得到證明，其值較低，並搭配低於 0.05 的顯著性。

Therefore, it complies with the multinomial regression model.

▶ 中文譯文 因此，它符合多項式迴歸模型。

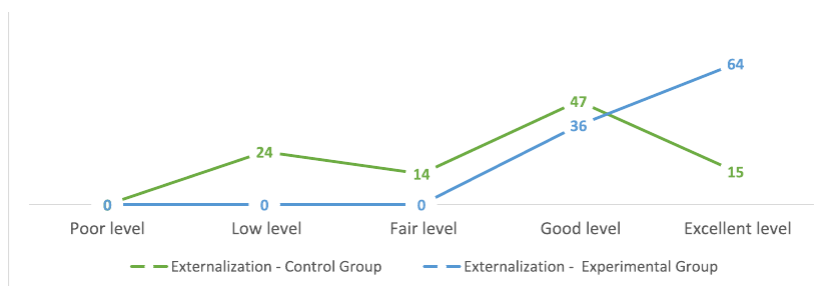


Fig. 5. Level of externalization.

Fig. 5. Level of externalization.

圖 5. 外化的水準。

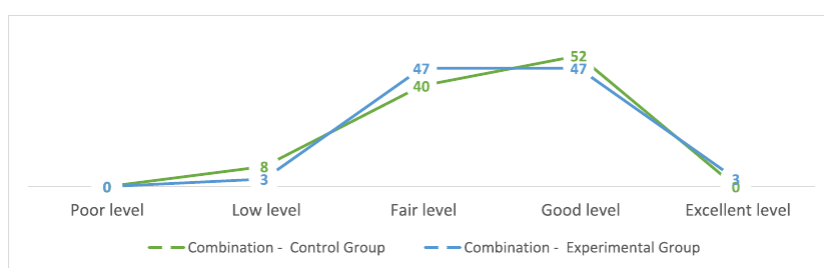


Fig. 6. Level of combination.

Fig. 6. Level of combination.

圖 6. 結合的水準。

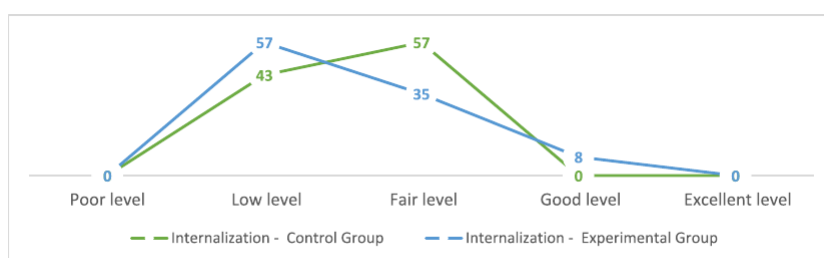


Fig. 7. Level of internalization.

Fig. 7. Level of internalization.

圖 7. 內化的水準。

表 5. 模型適配資訊－控制組。 Table 5. Model fit information – control group.

模型 Model	AIC	常態化 Normalized	對數概似 Log Likelihood 2	卡方 Chi-square	df	顯著性 Sig.
僅截距 Only Intercept	118.48	121.553	116.948			
最終	70.436	93.883	52.436	64.512	8	<0.001

模型 Model	AIC	常態化 Normalized	對數概似 Log Likelihood 2	卡方 Chi-square	df	顯著性 Sig.
Final						

In Table 9, the values obtained in the pseudo-R-squared are shown, selecting the highest, which is Nagelkerke's with a coefficient of 0.413.

► 中文譯文 表 9 呈現各項擬 R 平方的數值，取其中最高者，即 Nagelkerke 值，係數為 0.413。

This indicates that the final model explains 41.3 % of the variance in the dependent variable (internalization), which is a considerable value for measuring causality.

► 中文譯文 這代表最終模型能解釋依變項（內化）41.3%的變異，就衡量因果關係而言是個相當可觀的數值。

In Table 10, it can be seen that according to the results obtained, in the SECI model, the socialization factor predicts the internalization score because its p-value of 0.013 is statistically significant.

► 中文譯文 從表 10 可見，依所得結果，在 SECI 模型中，社會化因子能預測內化分數，因為其 p 值為 0.013，達到統計顯著。

Furthermore, the coefficient Exp (B) indicates that as socialization increases by one unit, the student is 0.156 times more likely to obtain a high score on the objective test conducted for internalization.

► 中文譯文 此外，Exp(B)係數顯示，社會化每增加一個單位，學生在內化客觀測驗中取得高分的可能性為 0.156 倍。

The externalization factor does not predict the internalization score owing to its p-value of 0.178, which is not statistically significant.

► 中文譯文 外化因子無法預測內化分數，因為其 p 值為 0.178，未達統計顯著。

Furthermore, the coefficient of Exp (B) or odds ratio indicates that as the externalization increases by one unit, the student is 3.330 times more likely to obtain a low score in the objective test conducted for internalization, as indicated by the negative beta coefficient.

► 中文譯文 此外，Exp(B)係數（即勝算比）顯示，外化每增加一個單位，學生在內化客觀測驗中取得低分的可能性為 3.330 倍，這也呼應了負的 beta 係數。

Similarly, the combination factor also does not predict the internalization score, as its p-value of 0.84 is not statistically significant.

► 中文譯文 同樣地，結合因子也無法預測內化分數，因為其 p 值為 0.84，未達統計顯著。

However, the coefficient Exp (B) shows that as externalization increases by one unit, the student is 1.165 times more likely to obtain a high score in the objective test conducted for internalization.

► 中文譯文 不過，Exp(B)係數顯示，外化每增加一個單位，學生在內化客觀測驗中取得高分的可能性為 1.165 倍。

註：a. 參考類別為「基礎」（Basic）。

Observing the results of the control and experimental groups, it is evident that the overall model does not improve with the incorporation of ChatGPT.

► 中文譯文 綜觀控制組與實驗組的結果，可以明顯看出整體模型並未因納入 ChatGPT 而改善。

This is demonstrated by comparing the pseudo-squared results with the SECI model without the use of ChatGPT, achieving 63.4 % predictability, while the SECI model with the use of ChatGPT achieves 41.3 % predictability.

► 中文譯文 這可由擬 R 平方的比較得到證明：未使用 ChatGPT 的 SECI 模型達到 63.4%的可預測性，而使用 ChatGPT 的 SECI 模型只達到 41.3%。

This implies that the use of AI does not contribute substantially to the increase of tacit knowledge in the internalization process.

► 中文譯文 這意味著，使用 AI 對於內化歷程中隱性知識的增長，並沒有實質性的貢獻。

However, a detailed analysis of each stage of the model reveals that in the externalization and combination phases, where the emphasis is on collecting theoretical information, the causal relation is positive with betas of 1.203 and 0.152, respectively, and significant in terms of the probability of improving the objective test with coefficients of 3.330 and 1.165.

► 中文譯文 然而，對模型各階段的細部分析顯示，在強調蒐集理論資訊的外化與結合階段，因果關係為正向，beta 值分別為 1.203 與 0.152，且就提升客觀測驗的機率而言具顯著性，係數分別為 3.330 與 1.165。

5. 討論 5. Discussion

Research on the impact of ChatGPT within the SECI model reveals mixed results, highlighting the utility of AI in certain phases of the learning process, while also emphasizing its limitations in others.

► 中文譯文 關於 ChatGPT 在 SECI 模型中影響的研究呈現出好壞參半的結果，一方面凸顯 AI 在學習歷程某些階段的效用，另一方面也強調它在其他階段的限制。

In the socialization phase, there was a notable improvement in interaction and student engagement in the experimental group, suggesting that ChatGPT facilitates idea generation and participation in collaborative discussions.

► 中文譯文 在社會化階段，實驗組在互動與學生投入上有顯著改善，顯示 ChatGPT 有助於想法產生與協作討論的參與。

The studies by Rahman and Watanobe [17] and Romero-Rodríguez et al. [9] support these findings, emphasizing that the innovative features of ChatGPT, such as personalized feedback and an accessible interface, motivate students to engage more actively in educational settings.

► 中文譯文 Rahman 與 Watanobe[17]以及 Romero-Rodríguez 等人[9]的研究支持這些發現，強調 ChatGPT 的創新特性（如個人化回饋與易於上手的介面）能促使學生在教育情境中更積極投入。

However, despite these initial benefits, prolonged use of this tool raises questions about how dependence on it could affect students' critical thinking and creativity [22].

► 中文譯文 然而，儘管有這些初期好處，長期使用這項工具仍引發疑慮：對它的依賴可能會如何影響學生的批判思考與創造力[22]。

表 6. 擬 R 平方－控制組。 Table 6. Pseudo-R-squared – control group.

指標 Index	數值 Value
Cox y Snell	0.475
Nagelkerke	0.634
McFadden	0.465

表 7. 參數估計－控制組。 Table 7. Parameter estimates – control group.

內化 Internalization	B	標準誤 Std. Error	Wald	df	顯著 Sig.	Exp(B)
中間 截距 Intermediate Intercept	-0.021	1.379	0.000	1	0.988	—
社會化 Socialization	0.032	0.302	0.011	1	0.916	1.032
外化 Externalization	1.249	0.333	14.107	1	0.000	3.488
結合 Combination	-1.312	0.480	7.471	1	0.006	0.269

In the externalization phase, the experimental group demonstrated an improvement in the quality of the outputs generated, indicating that ChatGPT facilitates the management of tacit knowledge, which is often challenging to articulate in collaborative activities [20].

► 中文譯文 在外化階段，實驗組在產出品質上有所提升，顯示 ChatGPT 有助於管理隱性知識，而這類知識在協作活動中往往難以清楚表達[20]。

Its real-time text generation capability also contributed to the originality and quality of creative solutions [19].

► 中文譯文 它即時生成文字的能力，也對創意解決方案的原創性與品質有所貢獻[19]。

However, concerns arise about the potential of overdependence on the tool, which could limit the development of critical skills essential for autonomous and creative learning.

► 中文譯文 然而，過度依賴這項工具的疑慮也隨之浮現，這可能會限制自主與創造性學習所不可或缺之批判能力的發展。

In the combination phase, the impact of ChatGPT was less marked.

► 中文譯文 在結合階段，ChatGPT 的影響較不明顯。

No notable improvement was observed in knowledge consolidation compared with the control group, suggesting that AI still faces limitations in providing consistent answers in complex subjects such as mathematics and exact sciences.

► 中文譯文 相較於控制組，知識整固上並未觀察到顯著改善，顯示 AI 在數學與精確科學等複雜學科中，提供一致答案的能力仍有限制。

Li et al. [24] noted that while ChatGPT could generate readable and useful texts for the dissemination of financial information, its accuracy in technical areas such as financial mathematics was inconsistent.

► 中文譯文 Li 等人[24]指出，ChatGPT 雖能為財務資訊的傳播生成可讀且有用的文本，但在財務數學這類技術領域的正確性並不穩定。

This highlights a critical limitation in using ChatGPT for tasks that require precision and a detailed understanding of numerical concepts, underscoring the need for other tools such as ComputeGPT to improve the accuracy in such activities.

► 中文譯文 這凸顯了在需要精確度與對數值概念細緻理解的任務上使用 ChatGPT 的一項關鍵限制，也突顯出需要 ComputeGPT 等其他工具來提升這類活動的正確性。

In the internalization phase, no considerable differences in knowledge retention were found between the experimental and control groups.

► 中文譯文 在內化階段，實驗組與控制組之間在知識保留上並未發現可觀的差異。

Indeed, the control group showed a slight advantage in retaining new knowledge.

► 中文譯文 事實上，控制組在保留新知識上還展現出些微的優勢。

This may be related to the lack of clear references provided by ChatGPT on specific topics such as financial mathematics, making it challenging to build new knowledge [25].

► 中文譯文 這或許與 ChatGPT 在財務數學等特定主題上未能提供清楚的參考依據有關，使得建構新知識變得困難[25]。

Despite this, the potential of ChatGPT as a pedagogical support tool for problem solving and lesson creation is acknowledged, although it still faces limitations in solving numerical problems.

► 中文譯文 儘管如此，ChatGPT 作為解題與課程設計之教學輔助工具的潛力仍受到肯定，只是它在解決數值問題上仍有限制。

According to this study, the overall impact of ChatGPT on the SECI model indicates that its contribution to the internalization process is limited, with a reduction in the predictability of the model from 63.4 % to 41.3 %.

► 中文譯文 根據本研究，ChatGPT 對 SECI 模型的整體影響顯示，它對內化歷程的貢獻有限，模型的可預測性從 63.4% 下降到 41.3%。

This suggests that, while ChatGPT facilitates phases such as socialization and externalization effectively, its use in the consolidation of tacit knowledge is less productive.

► 中文譯文 這顯示，ChatGPT 雖能有效促進社會化與外化等階段，但用於整固隱性知識時的成效較低。

This finding aligns with previous studies that have emphasized the need for human supervision and complementary pedagogical approaches to mitigate the potential negative consequences of AI dependency [26,44].

► 中文譯文 這項發現與先前研究一致，那些研究都強調需要人為監督與互補的教學取徑，以減緩對 AI 依賴可能帶來的負面後果[26,44]。

The research by Jensen et al. [15] presents an optimistic view of using ChatGPT in higher education, highlighting its ability to personalize and enhance learning assessments by integrating effectively into curricula as a complement to students' thinking and writing processes.

► 中文譯文 Jensen 等人[15]的研究對在高等教育中使用 ChatGPT 抱持樂觀看法，強調它能有效融入課程，作為學生思考與寫作歷程的補充，藉此個人化並強化學習評量。

While both studies acknowledge the educational utility of AI, this study takes a more cautious stance on its long-term impact on knowledge consolidation.

► 中文譯文 雖然兩項研究都肯定 AI 的教育效用，本研究對它在知識整固上的長期影響則採取較為審慎的立場。

Similarly, Saúde et al. [12] recognize the benefits of ChatGPT for academic performance and feedback; however, as in this study, they emphasize the need for pedagogical support to foster critical and ethical skills in students.

► 中文譯文 同樣地，Saúde 等人[12]肯定 ChatGPT 對學業表現與回饋的助益；不過，如同本研究，他們也強調需要教學上的支持，以培養學生的批判與倫理能力。

Likewise, Damaševičius [13] and the results of this study indicate that although ChatGPT can enhance certain educational activities, its limitations in contextual understanding, personalization, and the development of complex cognitive skills account for its reduced impact in advanced learning stages.

► 中文譯文 同樣地，Damaševičius[13]與本研究的結果都指出，ChatGPT 雖能強化某些教育活動，但它在脈絡理解、個人化以及複雜認知能力培養上的限制，正是它在進階學習階段影響力降低的原因。

This reinforces the idea that while ChatGPT is a valuable complement in some phases of the educational process, it cannot replace the depth of human learning.

► 中文譯文 這強化了一個觀點：ChatGPT 在教育歷程的某些階段是有價值的補充，但它無法取代人類學習的深度。

However, the students reported high levels of satisfaction and progress in the lessons wherein it was used, suggesting a positive effect associated with the novelty of the tool.

► 中文譯文 然而，學生在使用它的課堂中回報了高度的滿意與進步感，顯示出與這項工具新鮮感相關的正面效應。

The research by Leleparý et al. [10] highlights the use of ChatGPT in tasks such as translation and the development of reading skills through practical questions that promote critical thinking and collaboration.

► 中文譯文 Leleparý 等人[10]的研究凸顯了 ChatGPT 在翻譯等任務上的運用，並透過促進批判思考與協作的實務問題來培養閱讀能力。

This approach was well received by students, who appreciated the interactive environment facilitated by AI.

► 中文譯文 這種做法受到學生好評，他們很欣賞 AI 所營造的互動環境。

However, in this study, the limitations of ChatGPT in handling complex contexts and fostering deeper critical thinking are acknowledged.

► 中文譯文 然而，本研究也承認 ChatGPT 在處理複雜脈絡與培養更深層批判思考上的限制。

These findings emphasize that while useful in certain areas, its ability to support the internalization of knowledge is limited.

► 中文譯文 這些發現強調，它雖然在某些領域有用，但支援知識內化的能力是有限的。

Despite initial enthusiasm and its promotion of innovative practices, a cautious approach is needed to ensure the long-term reliability and effectiveness of ChatGPT as institutions address the challenges of its implementation.

► 中文譯文 儘管初期充滿熱情，並推動了創新做法，但在各機構面對導入挑戰之際，仍需以審慎的態度來確保 ChatGPT 長期的可靠性與成效。

The results of the study by Baidoo-Anu and Ansah [45] align with those of this study, highlighting the potential of ChatGPT to improve socialization and facilitate certain educational activities.

► 中文譯文 Baidoo-Anu 與 Ansah[45]的研究結果與本研究一致，凸顯了 ChatGPT 改善社會化、促進某些教育活動的潛力。

However, both studies emphasize the limitations of the tool in terms of deep learning, particularly in its precision and personalization, which is also reflected in the findings of this study on its limited impact on the combination and internalization of knowledge.

► 中文譯文 然而，兩項研究都強調這項工具在深度學習上的限制，尤其是在精確度與個人化方面，這也呼應了本研究關於它對知識結合與內化影響有限的發現。

The main difference between the studies lies in the approach: Baidoo-Anu and Ansah examine the use of ChatGPT in the education system in general, while this study focuses on an empirical analysis within the SECI model, explaining the identification of moderate and specific impacts in certain phases of collaborative learning.

► 中文譯文 兩項研究的主要差異在於取徑：Baidoo-Anu 與 Ansah 檢視的是教育體系中 ChatGPT 的一般性運用，而本研究聚焦於 SECI 模型內的實證分析，因而能辨識出協作學習特定階段中那些中等而具體的影響。

Meanwhile, a relevant aspect is highlighted in the research by Rudolph et al. [2], wherein the performance of several chatbots (including GPT-4, ChatGPT-3.5, Bing Chat, and Bard) in specific educational tasks was evaluated.

► 中文譯文 此外，Rudolph 等人[2]的研究凸顯了一個相關面向，他們評估了多款聊天機器人（包括 GPT-4、ChatGPT-3.5、Bing Chat 與 Bard）在特定教育任務上的表現。

While the usefulness of these chatbots is noted, their varying capabilities and limitations are also highlighted, making careful use essential in educational settings.

► 中文譯文 研究在肯定這些聊天機器人有用之餘，也凸顯了它們參差的能力與限制，使得在教育情境中謹慎使用顯得不可或缺。

表 8. 模型適配資訊－實驗組。 Table 8. Model fit information – experimental group.

模型 Model	AIC	常態化 AIC Normalized AIC	對數概似 Log Likelihood 2	卡方 Chi-square	df	顯著性 Sig.
僅截距 Only Intercept	109.378	114.589	105.378			
最終 Final	96.713	138.395	64.713	40.666	14	0.000

表 9. 擬 R 平方－實驗組。 Table 9. Pseudo-R-squared – experimental group.

指標 Index	數值 Value
Cox y Snell	0.344
Nagelkerke	0.413
McFadden	0.237

表 10. 參數估計－實驗組。 Table 10. Parameter estimates – experimental group.

內化 Internalization	B	標準誤 Std. Error	Wald	df	顯著 Sig.	Exp(B)
高 截距 High Intercept	2.278	5.391	0.178	1	0.673	—
社會化 Socialization	-1.859	0.748	6.178	1	0.013	0.156
外化 Externalization	1.203	0.892	1.817	1	0.178	3.330
結合 Combination	0.152	0.767	0.039	1	0.842	1.165

Similar to the findings of the present study, it is observed that the effectiveness of these tools is limited, whether in their ability to consolidate knowledge or due to their inconsistent performance and contextual errors.

► 中文譯文 與本研究的發現相似，他們觀察到這些工具的成效有限，無論是在整固知識的能力上，或是因其表現不穩定與脈絡上的錯誤所致。

The divergence between both studies can be attributed to different approaches; while the present research examines the impact of ChatGPT throughout the entire learning process within the SECI model, the study by Rudolph et al. focuses on the quality of responses from various chatbots on specific higher education tasks.

► 中文譯文 兩項研究的分歧可歸因於不同的取徑；本研究檢視的是 ChatGPT 在 SECI 模型整個學習歷程中的影響，而 Rudolph 等人的研究則聚焦於各款聊天機器人在特定高教任務上回應的品質。

In addition, Romero-Rodríguez et al. [9] present different findings, emphasizing the influence of habit and behavioral intention in the use of ChatGPT.

► 中文譯文 此外，Romero-Rodríguez 等人[9]提出了不同的發現，強調習慣與行為意圖在 ChatGPT 使用上的影響。

While our study shows a greater impact in the early phases of learning with reduced effectiveness in later stages, Romero-Rodríguez et al. focus on usage behaviors.

► 中文譯文 我們的研究顯示影響在學習的早期階段較大、後期階段成效下降，而 Romero-Rodríguez 等人則聚焦於使用行為。

Similarly, the research by Rezaev and Tregubova [1] aligns with the findings of this study, noting that ChatGPT enhances knowledge exchange and collaborative activities.

► 中文譯文 同樣地，Rezaev 與 Tregubova[1]的研究與本研究發現一致，指出 ChatGPT 能強化知識交流與協作活動。

However, while this study shows a limited impact in the advanced phases of the SECI model, such as combination and internalization, Rezaev and Tregubova emphasize the ability of ChatGPT to reduce task time, structure discussions more efficiently, and foster innovative solutions.

► 中文譯文 然而，本研究顯示在 SECI 模型的進階階段（如結合與內化）影響有限，Rezaev 與 Tregubova 則強調 ChatGPT 能縮短任務時間、更有效率地組織討論，並激發創新的解決方案。

These differences reflect variations in the depth of impact: while this work observes a moderate effect on deep learning, Rezaev and Tregubova highlight the practical use of ChatGPT in specific tasks and remote learning, suggesting that it enhances efficiency and creativity but has limited influence on knowledge consolidation.

► 中文譯文 這些差異反映出影響深度的不同：本研究觀察到對深度學習有中等程度的效果，Rezaev 與 Tregubova 則凸顯 ChatGPT 在特定任務與遠距學習上的實務運用，指出它能提升效率與創造力，但對知識整固的影響有限。

Likewise, Vicente-Yagüe-Jara et al. [46] support the findings of this study, indicating that ChatGPT has a positive impact on activities that require rapid idea generation, socialization, and problem solving.

► 中文譯文 同樣地，Vicente-Yagüe-Jara 等人[46]支持本研究的發現，指出 ChatGPT 對需要快速產生想法、社會化與解決問題的活動具有正面影響。

However, the divergence lies in the focus; this study centers on knowledge transfer and consolidation, whereas Vicente-Yagüe-Jara et al. evaluate creativity and its comparison between AI and humans.

► 中文譯文 然而分歧在於關注焦點；本研究以知識的傳遞與整固為核心，而 Vicente-Yagüe-Jara 等人則評估創造力，以及 AI 與人類之間的比較。

These differences may be attributed to the complexity of the tasks analyzed and the learning phases where AI is employed.

► 中文譯文 這些差異或許可歸因於所分析任務的複雜度，以及運用 AI 的學習階段不同。

Although ChatGPT facilitates certain processes, it continues to face challenges in tasks requiring deeper knowledge integration or higher levels of creativity.

► 中文譯文 儘管 ChatGPT 能促進某些歷程，它在需要更深層知識整合或更高層次創造力的任務上仍持續面臨挑戰。

The research by Qadhi et al. [11], although not directly assessing the impact of AI on collaborative learning or knowledge acquisition, provides a conceptual framework for its ethical implementation, emphasizing the importance of maintaining academic standards in higher education.

► 中文譯文 Qadhi 等人[11]的研究雖未直接評估 AI 對協作學習或知識習得的影響，卻為其倫理性導入提供了概念架構，強調在高等教育中維持學術標準的重要性。

Similarly, Akpan et al. [14] take a global perspective, analyzing how generative AI technologies are rapidly transforming education and research, but without exploring the quality of knowledge acquisition, as this study does with the SECI model.

► 中文譯文 同樣地，Akpan 等人[14]採取全球視角，分析生成式 AI 技術如何快速地改變教育與研究，但並未像本研究透過 SECI 模型那樣探討知識習得的品質。

Finally, the SECI model and ChatGPT could be optimized with insights from Samsuryadi [6], who proposes how advanced technologies can improve curriculum design and better prepare students for the labor market, offering a structural approach to enhance the efficiency of the model.

► 中文譯文 最後，SECI 模型與 ChatGPT 還可借助 Samsuryadi[6]的洞見加以優化，他提出先進技術如何改善課程設計、讓學生更能為勞動市場做好準備，為提升模型效率提供了一個結構性的取徑。

6. 結論 6. Conclusions

The results revealed that although the use of ChatGPT improved the quality of interaction and externalization of knowledge in the experimental group compared to the control group, it had no notable impact on the quality of combination or internalization of knowledge.

► 中文譯文 結果顯示，相較於控制組，使用 ChatGPT 雖然提升了實驗組在互動與知識外化上的品質，卻對知識結合與內化的品質沒有顯著影響。

Furthermore, multinomial regression analysis indicated that the inclusion of ChatGPT in the SECI model did not improve its predictability.

► 中文譯文 此外，多項式迴歸分析指出，在 SECI 模型中納入 ChatGPT 並未提升模型的可預測性。

That is, the use of ChatGPT did not substantially enhance the SECI model, as evidenced by the decrease in the pseudo-R-squared in both groups (63.4 % without ChatGPT compared to 42.2 % with ChatGPT).

► 中文譯文 也就是說，使用 ChatGPT 並未實質強化 SECI 模型，這由兩組擬 R 平方的下降得到證明（未使用 ChatGPT 為 63.4%，使用 ChatGPT 則為 42.2%）。

Results indicate that ChatGPT can facilitate tacit knowledge management and enhance expression during collaborative activities; however, it has limitations in mathematical comprehension and the promotion of critical skills.

► 中文譯文 結果顯示，ChatGPT 能促進隱性知識的管理，並在協作活動中強化表達；然而，它在數學理解與批判能力的培養上有所限制。

Although AI is essential, difficulties in obtaining precise information and performing specialized numerical operations require careful consideration when implementing these technologies.

► 中文譯文 儘管 AI 不可或缺，但在取得精確資訊與執行專業數值運算上的困難，使得導入這些技術時必須審慎考量。

The research also raises concerns about the reliability of the information provided by AI tools and the need for human supervision to ensure the quality of the content.

► 中文譯文 本研究也對 AI 工具所提供資訊的可靠性，以及為確保內容品質所需的人為監督，提出了關切。

Furthermore, it highlights the lack of a single tool to address all aspects of learning.

► 中文譯文 此外，它也凸顯出沒有單一工具能涵蓋學習的所有面向。

Ultimately, the use of ChatGPT does not contribute substantially to tacit knowledge according to the SECI model, indicating that its use does not improve the predictability of the model.

► 中文譯文 歸根究柢，依 SECI 模型來看，使用 ChatGPT 對隱性知識並無實質貢獻，顯示它的使用並未提升模型的可預測性。

These findings underscore the need to understand the limitations and challenges when implementing technologies such as ChatGPT in educational settings.

► 中文譯文 這些發現凸顯出，在教育情境中導入 ChatGPT 這類技術時，必須理解其限制與挑戰。

Valuable information is provided on the impact of ChatGPT on learning, highlighting the need for future research to explore its effective integration in different educational contexts and disciplines, and to better understand its limitations and potential.

► 中文譯文 本研究就 ChatGPT 對學習的影響提供了寶貴的資訊，並凸顯未來研究有必要探討如何將它有效整合到不同的教育情境與學科中，以更深入理解它的限制與潛力。

CRediT 作者貢獻聲明 CRediT authorship contribution statement

Urpi Barreto: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization.

► 中文譯文 Urpi Barreto：撰寫－審閱與編修、撰寫－初稿、驗證、研究方法、形式分析、概念構想。

Yasser Abarca: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization.

► 中文譯文 Yasser Abarca：撰寫－初稿、調查執行、形式分析、資料整理、概念構想。

資料可取得性聲明 Data availability statement

The data will be available upon request.

► 中文譯文 資料可依需求提供。

倫理核准 Ethical approval

The study, being educational and non-invasive in nature, was conducted in full compliance with applicable ethical standards.

► 中文譯文 本研究本質上屬教育性且非侵入性，全程遵循適用的倫理標準執行。

Informed consent was obtained from all participants in accordance with established ethical principles and research regulations.

► 中文譯文 已依據既定的倫理原則與研究規範，取得所有參與者的知情同意。

This consent was secured through a transparent process that ensured respect for participants' rights, the anonymization of their data, and their entirely voluntary participation.

► 中文譯文 該同意是透過一套透明的程序取得，確保尊重參與者的權利、其資料的匿名化，以及他們完全出於自願的參與。

利益衝突聲明 Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

► 中文譯文 作者聲明，他們沒有任何已知、可能影響本論文所述研究的競爭性財務利益或個人關係。

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